A First Look at the Name Resolution Latency on Handshake

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ABSTRACT

The domain name system (DNS) has been an important base of the Internet; however, it faces various issues, including DNS server attacks. Blockchain-based DNS services have recently emerged to solve these DNS issues. Among them, Handshake provides a decentralized root zone and supports state-of-the-art features. However, few studies have investigated the performance evaluations of Handshake. We measure the name resolution latency on Handshake to assess its suitability as an alternative to the current DNS root servers and reveal that it has a practical drawback compared to current root servers at the current early stage.

CCS CONCEPTS

• **Networks** → **Naming and addressing**; **Network measurement**; *Peer-to-peer protocols*;

KEYWORDS

Blockchain, Blockchain-based domain name system, Domain name system

ACM Reference Format:
Katsuki Isobe, Daishi Kondo, and Hideki Tode. 2022. A First Look at the Name Resolution Latency on Handshake. In *Proceedings of the 22nd ACM Internet Measurement Conference (IMC '22), October 25–27, 2022, Nice, France.* ACM, New York, NY, USA, 2 pages. https://doi.org/10.1145/3517745.3563024

1 INTRODUCTION

The domain name system (DNS) faces various problems, including security and privacy, owing to its architecture [3]. Blockchain has garnered considerable attention, and there have been attempts recently to utilize its benefits to address these issues [1]. Handshake[1] aims to resolve the centralization of the top-level domain (TLD) by ICANN[2] by replacing the current DNS root servers with blockchain nodes. Owing to the decentralized nature of blockchain, Handshake makes TLD registration more resilient to censorship. Additionally, Handshake secures DNS records more easily with lower installation cost[3] of DNS security extensions than the current DNS. On Handshake, any string can be used as Handshake TLDs, except for some special letters. Unlimited TLDs can be obtained by winning automated auctions or buying names from marketplaces, except for names reserved for other entities, to avoid name collisions (e.g., TLDs in the current DNS).

Handshake comprises two types of nodes: *full node* and *light node*[4]. The *full node* stores all the blockchain data, whereas the *light node* stores only the block headers. However, the *light node* ensures data integrity as a simplified payment verification node. Popular *full client* and *light client* implementations are *hsd* and *hnsd*, respectively. When a user utilizes a *full node* or *light node*, the user's node first attempts to connect to *full nodes*, whose addresses are already hardcoded in the program. These nodes are called *seed nodes* and are managed by Handshake communities. After the user's node connects to the *seed nodes*, the addresses of the other nodes are introduced to the user's node. Thereafter, it can communicate with the nodes introduced in the blockchain network as its peers. Additionally, client programs have the option to change the maximum number of simultaneous connections, which is called *pool size*. Furthermore, Handshake supports two types of P2P connection: *clearnet* and *brontide*. The *clearnet* is an unencrypted connection between peers, whereas the *brontide* is encrypted. The *brontide* is essentially used only in communication between *seed nodes* or between *seed node* and the other.

Regarding the name resolution[5], the *full client* can resolve the Handshake TLDs locally and can perform a usual name resolution with the recursive resolver that runs in the *full client* after the Handshake TLD resolution. The *light client* can resolve the Handshake TLDs by making requests to the connected *full nodes* for proof data including DNS records in the blockchain. The *light node* maintains the pool of peers (only *full nodes*) limited by *pool size* and performs name resolution to peers randomly selected from the peer pool. After the TLD resolution, the *light client* performs further name resolution the same as that of the *full client*.

Latency is an important DNS measure for the DNS performance; therefore, the Handshake metrics should be evaluated. Here, we focus on *hnsd* as the *light client* because the *light client* is more suitable for general use than the *full client*. We assess the suitability of Handshake by measuring the latency on Handshake and the current DNS root servers.

2 EXPERIMENTS

We prepared eight *light clients* (*hnsd*[6]) running in AWS EC2 instances[7] located in AU, BH, BR, DE, JP, US (east), US (west), and ZA[8].

[1]Available at https://handshake.org.
[2]Available at https://www.icann.org.
[3]Small fees are required to update the data on blockchain. Available at https://learn.namebase.io/about-handshake/mining-hns.

[4]Hereinafter, we call a blockchain node *full node* or *light node* when we focus on the blockchain network while we call a node *full client* or *light client* when we focus on node implementation.
[5]Hereinafter, we call receiving records (e.g., glue records) from Handshake blockchain Handshake TLD resolution or name resolution.
[6]We used the *hnsd* committed on January 4, 2022. Available at https://github.com/handshake-org/hnsd/tree/b3c85c8f42cb1dd615df720e0e16a9f7a8de8fe2.
[7]t3.micro (vCPUs: 2, memory: 1GiB, network burst bandwidth: 5Gbps).
[8]These codes represent Australia, Bahrain, Brazil, Germany, Japan, the United States (east), the United States (west), and South Africa, respectively.

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IMC '22, October 25–27, 2022, Nice, France
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ACM ISBN 978-1-4503-9259-4/22/10.
https://doi.org/10.1145/3517745.3563024

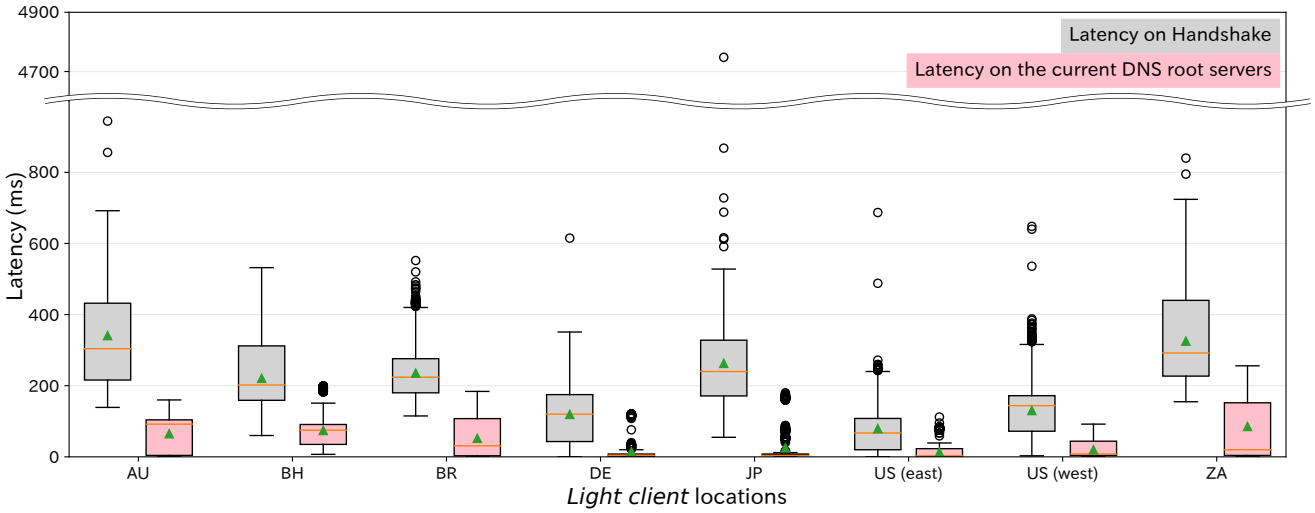

Figure 1: Name resolution latency on Handshake and the current DNS root servers.

First, we retrieved the name datasets of Handshake and the current DNS root servers from the volunteer's website[9] of the Handshake communities and Tranco dataset[10] [2], respectively. Thereafter, we created the TLD datasets to query by randomly selecting 100 TLDs for each client from the former dataset and extracting unique top-100 TLDs from the latter. The experiments for Handshake[11], each client sent 100 queries in each *pool size* option at intervals of 30 s to measure the latency. The *pool size* varied from the default value of 8–80 in eight increments. In the experiments for the current DNS root servers[12], each client sent 100 × 13 (A~M root letters) queries at the same interval to measure the latency of each root letter. The intervals were set so as not to adversely affect the entire network, that is, to avoid excessive overload.

Figure 1 depicts the name resolution latency on Handshake and the current DNS root servers per location. The green triangle and orange lines indicate the average and median values, respectively. Note that we only adopted responses whose `RCODE` was `NOERROR` to calculate the latency. Latency is significantly influenced by location. The average on Handshake is approximately six times as large as that on current DNS root servers. The total number of *full nodes* that responded to our clients during name resolution was 34. Specifically, 19, 7, 4, and 4 nodes were in the U.S., China, Germany, and other countries, respectively. Among them, 8 nodes were *seed nodes* that used *brontide*. Based on our investigation, we observed that the *full nodes* returning responses were not geographically distributed. This demonstrates that the latency decreases if the *light node* is geographically close to the *full nodes*; otherwise, the latency increases.

We subsequently examined whether the latency was changed by the difference in *pool size*. We observed that the communication ratio with the non-*seed nodes* to that with the *seed nodes* increased as the *pool size* increased (Figure 2). Furthermore, the latency with *brontide* was more than 1.6 times as high as that with *clearnet*, indicating that the encrypted connection might cause overhead.

[9]Available at http://adboard.durendil.hns.to (accessed January 30, 2022).
[10]Available at https://tranco-list.eu/list/2369/1000000 (accessed January 14, 2022).
[11]The experiments were held on March 9, 2022.
[12]The experiments were held on February 3, 2022.

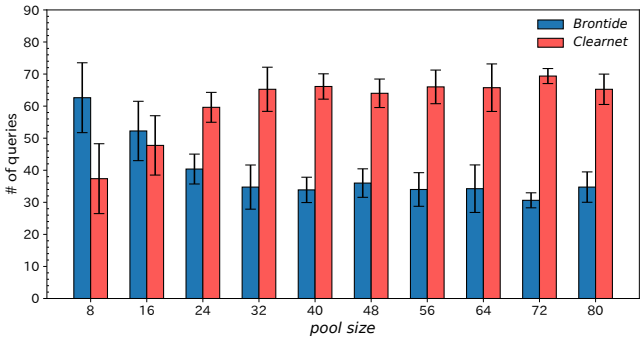

Figure 2: Comparison of the number of queries sent as the proof requests between *brontide* and *cleanet* per *pool size* (error bar: standard deviation)

The average latency decreased when the nodes increased the *pool size* because *brontide* was used only in the communication between the *seed node* and *light node*, as described in Section 1.

Services hosted with Handshake are negatively affected by the large latency. Therefore, at the current early stage, a practical drawback exists in Handshake against the current DNS root servers regarding latency.

Future research will include a detailed analysis of the cause of the high latency (e.g., the *brontide* overhead) and a proposal to reduce the latency on Handshake to the same level as the current DNS root servers.

ACKNOWLEDGEMENT

This work was supported by JSPS KAKENHI Grant Number JP21K17741.

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